

DAMAVARAPU DEVA HARSHA

ddharsha16@gmail.com

Location : Nellore,AP,India

Mob No: 8985556699

LinkedIn URL: <https://www.linkedin.com/in/devaharshadamavarapu/>

Github URL : <https://github.com/devaharsha23>

EDUCATION

Koneru Lakshmaiah Educational Foundation, Guntur	2022-2025
<i>Bachelor of Technology in Electronics and Communication Engineering (Specialization in VLSI)</i>	CGPA: 9.88
Sree Vidyanikethan Engineering College, Tirupati	2019-2022
<i>Diploma in Electronics and Communication Engineering -(S.B.T.E.T)</i>	Percentage: 97.77
Vowel School, Nellore	2018-2019
<i>Board of Secondary Education Andhra Pradesh (B.S.E.A.P)</i>	CGPA: 10

TECHNICAL SKILLS

Programming Languages: C, Embedded C, Python

HDL & Verification: Verilog, SystemVerilog, UVM (basic testbench development)

EDA Tools: Cadence Virtuoso, Xilinx Vivado, Intel Quartus Prime, ModelSim, MATLAB

Design & Simulation Tools: LTSpice, NI LabVIEW, Proteus

PCB Design Tools: KiCad, EasyEDA

Operating Systems: Linux (Ubuntu), Basic Kernel Understanding

Microcontrollers: Arduino, ESP32, NodeMCU, STM32, PIC, Raspberry Pi (Pico, Zero, 3, 4)

Test & Measurement: CRO, Function Generator, Spectrum Analyzer

Relevant Coursework: VLSI Design, Analog Circuit Design, Digital Design, Embedded Systems, Electronic Devices & Circuits

INTERNSHIP EXPERIENCE

INDIAN INSTITUTE OF SCIENCE (IISc) BANGALORE

Mar2025-June2025

RESEARCH INTERN

- Conducted research on Integrated Sensing and Communication (ISAC) for next-generation 6G systems, focusing on joint communication-radar paradigms
- Developed and implemented joint signal detection and target estimation algorithms in MATLAB, enabling simultaneous sensing and data transmission
- Applied projection-based interference mitigation techniques to suppress uplink interference and enhance detection accuracy

SDSC-SHAR ISRO

Mar2022-May2022

TRAINEE

- Acquired hands-on experience in PCB design, fabrication, and soldering for aerospace-grade electronic systems
- Developed and tested Arduino-based communication modules for telemetry applications and tracking mechanisms
- Contributed to balloon tracking and radar communication system testing, gaining exposure to real-time aerospace communication workflows

PROJECTS

HIGH ALTITUDE BALLOON -STUDENT MADE SATELLITE (KLSAT)

- Designed and launched a high-altitude balloon satellite reaching 28 km altitude and ~70 km range
- Worked on microcontrollers, Linux systems, and camera module integration
- Designed HF/VHF antennas for communication
- Implemented GPS-based tracking for real-time monitoring
- Performed PCB design and soldering for onboard systems

AHB2APB BRIDGE DESIGN

- Designed RTL for AMBA AHB-APB bridge enabling protocol conversion
- Implemented AHB slave & APB master interfaces with address/control handling
- Developed FSM-based control logic (IDLE, SETUP, ENABLE)
- Supported read/write transactions with proper handshake signals (HREADY, PENABLE, PREADY)
- Verified using SystemVerilog testbench and simulation in ModelSim/Vivado

HIGH LINEARITY MIXER DESIGN USING 180nm CMOS TECHNOLOGY

- Designed a CMOS RF mixer employing multi-tanh transconductance linearization to suppress third-order intermodulation distortion (IMD₃)
- Achieved improved linearity performance (IIP₃ enhancement) while maintaining competitive conversion gain and power consumption
- Optimized transistor sizing, biasing conditions, and load configuration to balance linearity, noise figure, and gain
- Performed circuit-level simulations including DC, AC, transient, and PSS/PAC analyses using Cadence Virtuoso
- Evaluated key RF metrics: IIP₃, P_{1dB}, conversion gain, noise figure, and port-to-port isolation

TRAINING AND CERTIFICATIONS

1. VLSI Design	July 2023 Virtual
2. Embedded Systems	May 2023 Virtual
3. Linux Administrator	May 2024 virtual
4. NI CLAD Lab View Developer	Feb 2023 virtual
5. VLSI design and verification	Feb 2024 virtual
6. India Ham Radio Certification and License	May 2024 Virtual

POSITIONS OF RESPONSIBILITIES

1. Communication Engineer & Team Lead- KLSAT (Satellite Club) – KL University	June 2023 -2025
2. Member of IEEE Digital Affairs Team [DAT] IEEE MTT-S Region10	Jan2025 – Dec 2025
3. ChairPerson IEEE MTT Society - KL University, Guntur	Dec 2022 – May2025
4. Member @Always VLSI Club – KL University	Dec 2022-Apr2023

AWARDS

1. **Silver Medalist** in **B.Tech** (Electronics & Communication Engineering)
2. ISWDP Samsung Student Fellowship Awardee -2025
3. Winner, Intel Technical Event
4. **Gold Medalist** in **Diploma** (Electronics & Communication Engineering) – 2019–2022
5. Recipient of the IEEE Best Student Mentee Award – 2023

PUBLICATIONS / RESEARCH WORK

Raspberry Pi-Based High-Altitude Balloon Satellite (KLSAT) for Live Image Capture and SSTV Downlink (Manuscript under review)

- Designed and developed a Raspberry Pi based high-altitude balloon satellite for real-time image capture and SSTV transmission.
- Implemented VHF communication link for telemetry and image downlink.
- Integrated GPS tracking, camera module, and RF antenna system.
- Achieved **28 km vertical altitude and ~70 km horizontal flight distance** during experimental launch.

Projection-Based Interference Mitigation for Uplink ISAC Systems (Research Work, IISc Bangalore)

- Analyzed radar–communication interference in uplink ISAC systems
- Developed projection-based linear detection algorithm for joint signal detection and target estimation
- Evaluated performance using BER, detection probability, and interference suppression metrics
- Demonstrated improved BER and interference resilience over SIC-based detection
- Validated scalability with receive antennas (M_r) and targets (L) for JRC applications

RESEARCH INTERESTS

- Analog & Mixed-Signal IC Design
- RF Circuit Design and Communication Systems
- Integrated Sensing and Communication (ISAC)
- ASIC Design for Sensing and Signal Processing Applications
- Hardware–Algorithm Co-Design for Communication Systems